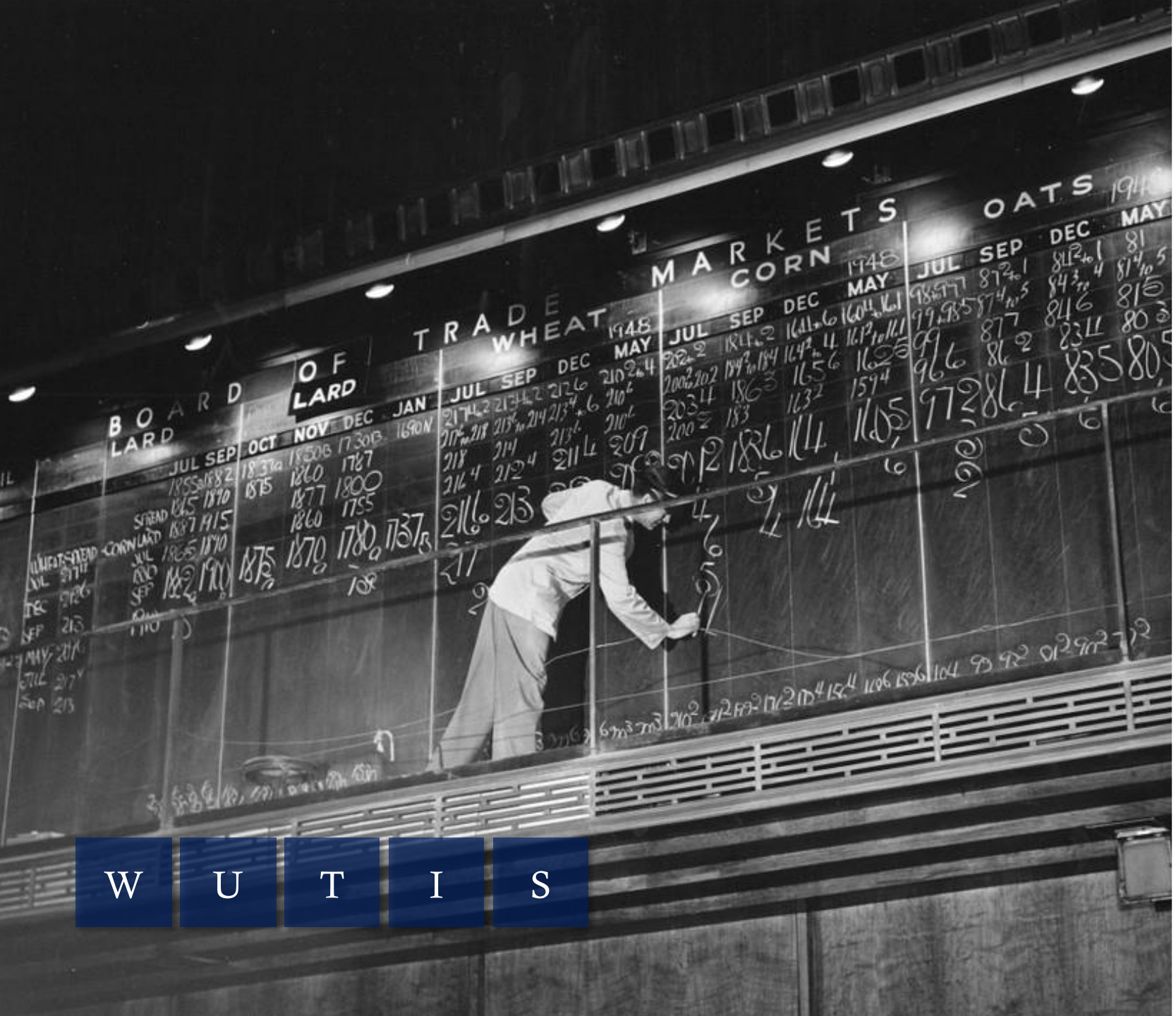


W U T I S

Global Markets Division

Trading Dispersion

Inspecting volatility regimes, rethinking a broken correlation game and reviving the hit of 2000s.

Next Month Trade:

Long Dispersion (EPS Growth 12M)

Matvii Polishchuk, Zhangir Mukhamejan, Renata Cheremnykh, Moritz Ransmayr, Kiril Zaklunnyi

Vienna, 5th May 2026

Team Overview

Global Markets



Matvii Polishchuk

Senior Associate

- Storyline
- Trading Strategy

Raiffeisen Bank International

BSc. (WU) – 4th Sem.

Pictures & Names Deleted Due to Privacy Concerns
Contact privately for more details.

Associate

- Structured Products

BSc. (WU) – 4th Sem.

Fellow Analyst

- Market Analysis

MSc. (UniWien) – 1st Sem.

Fellow Analyst

- Market Analysis

BSc. (WU) – 4th Sem.

Fellow Analyst

- Regime Analysis

BSc. (WU) – 4th Sem.

© WUTIS – Global Markets

Agenda

1	Structured Product Dynamics	3
2	Market Analysis: Monetary and Fiscal Regime	6
3	Market Analysis: Metals Flows & Correlations	10
4	Regime Analysis: Hidden Markov Model	13
5	Introduction to Dispersion Trading	17
6	Trading Strategy	19

© WUTIS – Global Markets



Structured Products

Rise of Structured Products

Structured Products Became Large Enough To Matter

How are They "Structured"?

Product Structure
Bond-like **funding plus embedded options** on indices, baskets, or single stocks.

Investor Objective
Investors typically buy them for **higher yield**, **downside protection**, or **conditional upside**.

Market Impact
When dealers sell them, they inherit **non-linear risk** that must be hedged in **listed markets**.

Main types

Reverse Convertible Notes
- High coupon
- Investor takes **downside risk**
- Dealer gets **downside option exposure**

Principal-protected Notes
- Capital protection
- Conditional upside
- Dealer hedges **long-dated exposure**

Worst-of Autocallables
- Enhanced yield in stable markets
- Barrier / observation-date sensitive
- **Highly non-linear** dealer exposure

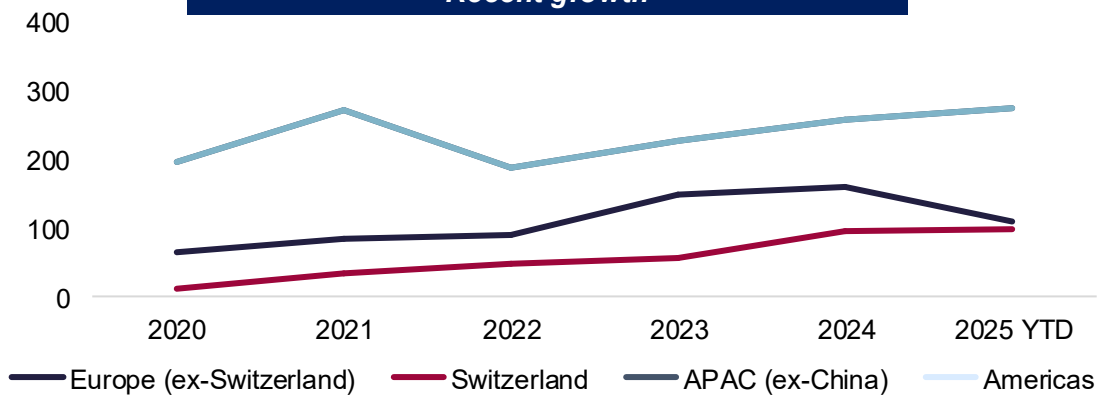
From Geek to Greek

From Niche Product to Market-relevant Flow

Historically	*Recent Shift*	*Implication*
▪ Introduced in **Europe** ▪ Sold mainly to **wealth / retail investors** ▪ Limited market **footprint**	▪ Issuance **accelerated** after 2020 ▪ Post-2022 **demand strengthened** ▪ Market reached **material scale**	▪ Larger **dealer books** ▪ More persistent **hedging** ▪ Greater **market relevance**

Larger issuance means larger **dealer books** and more persistent **hedge flows.**

Recent growth



Source: Structured Retail Products

© WUTIS – Global Markets

The Importance of Structured Products for Volatility

Dealer Hedging is the Channel Through Which Structured Products Affect Volatility

How Does the Effect Reach the Market?

1. **Risk-seeking Demand**
   Investors get exposure to the convex, high return opportunities.
2. **Selling the Risk**
   Structured products are issued to insure that growing demand.
3. **Exposure of Market Makers**
   Dealer inherits convexity in volatility & delta exposure.
4. **Premium Coverage**
   Dealer need to remain delta hedged in listed futures and options.
5. **Markets Touchpoint**
   Self-amplifying flows significantly change the reaction.

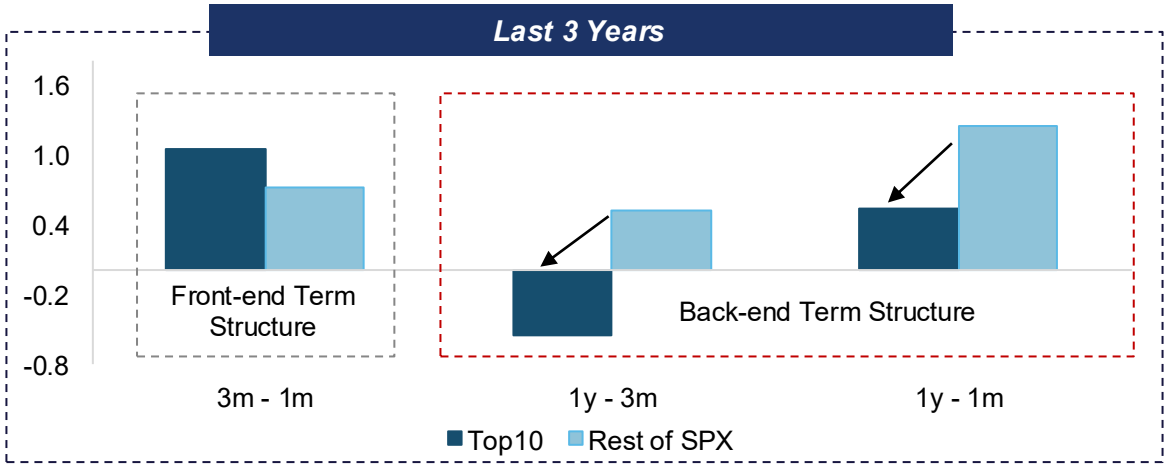
Observed Effect in Volatility

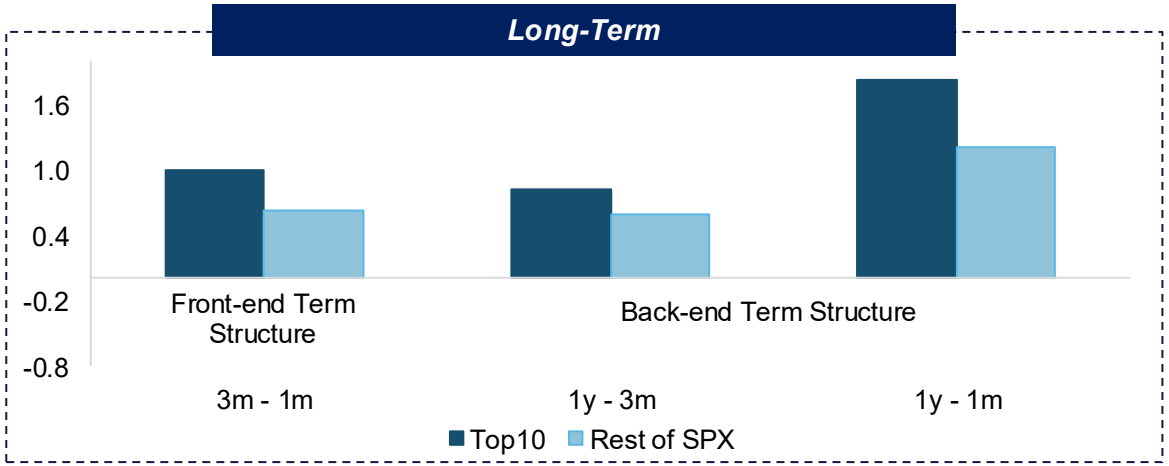
Recent period
- Top **autocall** names show lower **long-dated vol** than the rest of SPX.
- The effect is strongest in the **back end.**
- This is consistent with **dealer re-hedging.**

History
- The pattern was **weaker** in the past.
- That points to a **recent issuance-driven distortion.**

Takeaway
Structured products can affect **volatility pricing itself**, pointing to the new regime identification.

Structured Products Already Affect Volatility





Source: Barclays Derivatives Research

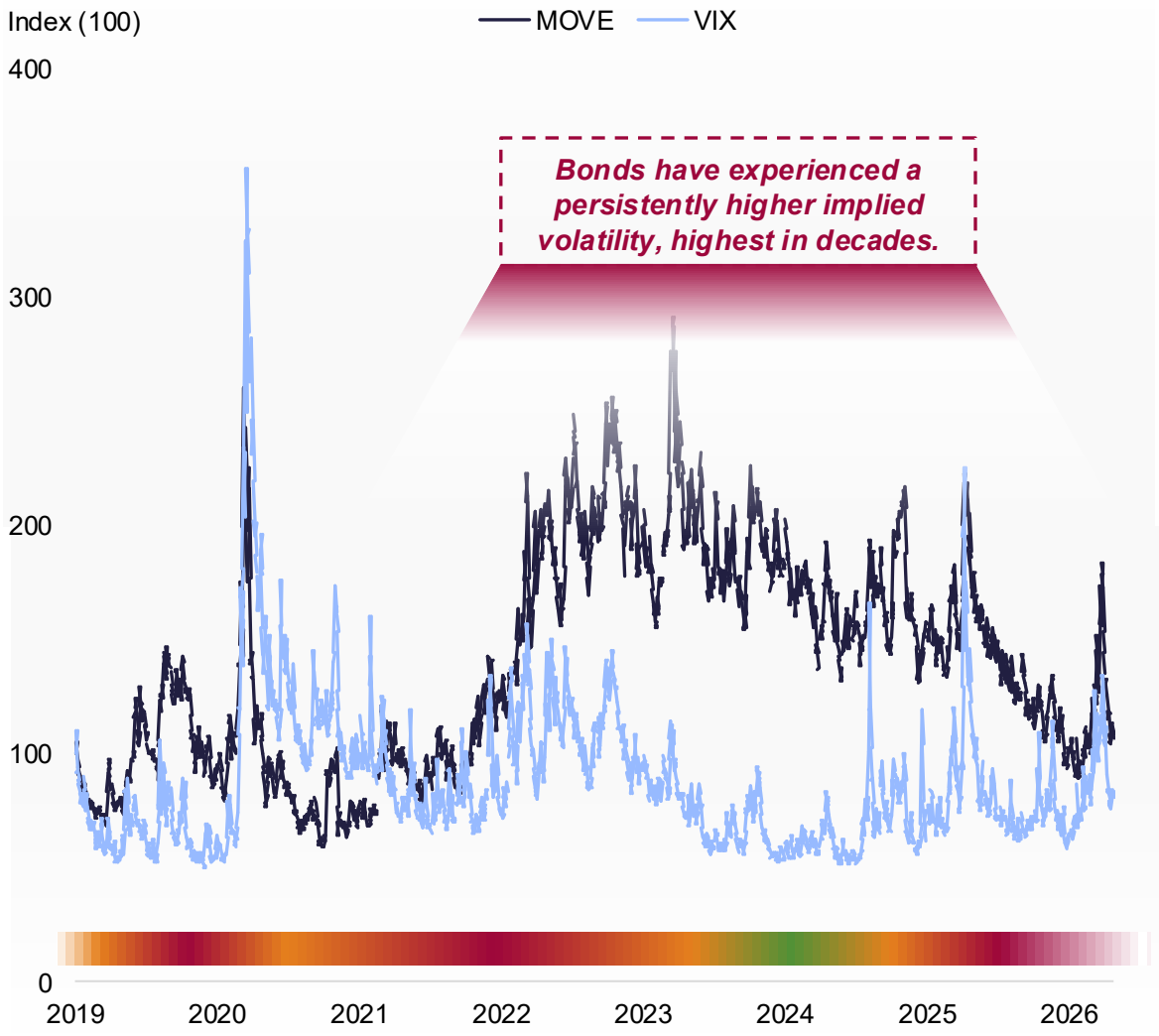
© WUTIS – Global Markets



Market Analysis

Bonds Perception and Historical Narrative

Through The Lense of Bond and Stock Volatility Index (MOVE & VIX)



Apr 2018 – Jan 2020

Policy Pivot and Bonds Uncertainty
Late-cycle Fed tightening lifted rate uncertainty and kept bond volatility elevated.

Feb 2020 – Dec 2021

COVID Crisis and Money Emission
FED cuts interest rates to zero, launches unlimited QE and provides massive fiscal stimulus, which caused a shift to supply-constrained economy.

Jan 2022 – Feb 2023

Lost Trust in Bonds and Inflation Shock
CPI peaked above 9%, forcing the fastest hiking cycle in decades, after which both stocks and bonds sold off together, breaking the 60/40 framework.

Mar 2023 – Aug 2024

Higher for Longer: Bond's Perception Changed
SVB and regional banks failures exposed the lost trust in the bonds market, while also pricing the increasing fiscal risk.

Sep 2024 – Dec 2025

Trump: Cuts and Tariffs
Fed began cutting as inflation eased and growth slowed, however, long yields rose instead of falling indicating a new regime.

Jan 2026 – Present

Fiscal Dominance and Inflation Fears
Currently bond supply and fiscal deterioration remain the key drivers of the bond market, while also pricing increased geopolitical risk and rate uncertainty.

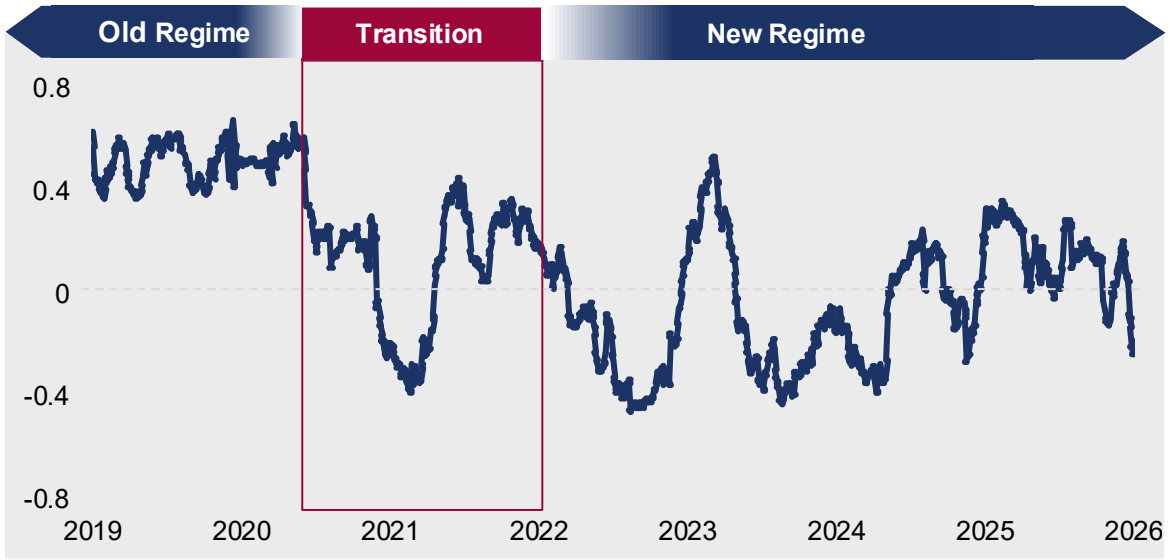
Source: Yahoo Finance

© WUTIS – Global Markets

Breakage of the 60/40 Portfolio

The Shift of Stock-Bond Relationship

US10Y and S&P Returns: 3M Rolling Correlation



Positive Correlation

Persistently elevated yields were associated with the **higher growth pricing** before the pandemics.

Sharp Breakdown

Yields decoupled during the crisis, due to rapid QE and **risk premia fears overriding the market decline**.

Unstable Correlation

No classic hand-book stability, **correlation is mean-reverting**, oscillating **around zero**.

Old Regime – Why Did it Work?

Low Inflation and Stable Rates
Environment created a low-vol macro backdrop for bonds to rally.

Fed Easing in Risk-off Episodes
Trust in regulator supported bond prices exactly when equities sold-off.

Bonds As Income & Hedge
Bonds delivered stable positive returns with structural diversification.

Structural break in 2022 as inflation and unstable FED policy caused both bonds and stocks to sell off simultaneously.

The New Regime – Why Diversification Broke?

Volatile Correlation

Bond-Equity **correlation** is currently **regime dependent** and volatile.

No Diversification

The **60/40** structure **no** longer guarantees **draw-down protection** in risk-off episodes.

Portfolio Implication

Need for **alternative hedges surges** across portfolio manager sentiment.

The 60/40 framework is no longer sufficient in a regime-driven market and diversification must be actively managed.

Sources: FRED, Yahoo Finance

© WUTIS – Global Markets

Current Macro Narrative and Policy Overview

Fiscal Dominance and a Regime-Driven Market

U.S. Fiscal Deterioration: Persistent Deficit and Debt Pressure

Annual Budget Deficit	US Government Debt	Annual Interest Expense
$1.7–2tn	$39tn	$1tn

Structural Treasury Supply Pressure
Excess bond supply weakens price support and increases yield volatility.

Macro-Defined Markets – What Drives Volatility and Correlation Breaks?

Fiscal Supply
Record Treasury issuance keeps upward pressure on long-end yields.

Geopolitics
Energy shocks and global tensions raise inflation risk premium.

Policy Uncertainty
FED path, tariffs and elections increase cross-asset volatility.

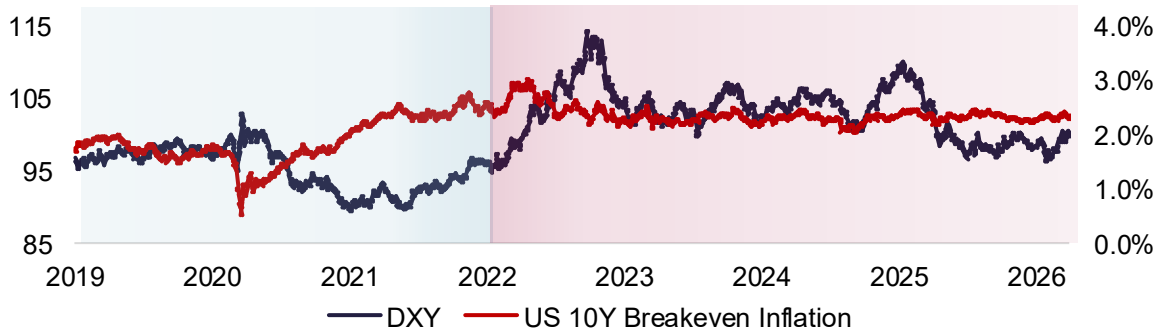
Repricing Role of Bonds

Treasuries are No Longer a Reliable Hedge
Bonds increasingly sell off alongside equities during stress episodes, and 60/40 dynamic is broken.

Yields Are Driven Beyond Monetary Policy
Fiscal dynamics and inflation now play a key role in shaping yields and bonds' volatility.

Bonds Behave as a Macro Asset
Elevated volatility and shifting cross-asset correlations limit their hedging role.

U.S. Dollar and Inflation Expectations



The new regime has driven macro-defined markets, reshaping various cross-asset dynamics and correlations.

Sources: FRED, Yahoo Finance

© WUTIS – Global Markets

Global ETF Flows Into Precious and Industrial Metals

The 60/40 Obituary Sent Capital Hunting For a New Home

Gold ETF Holdings by Region (Tonnes)

Record Inflows

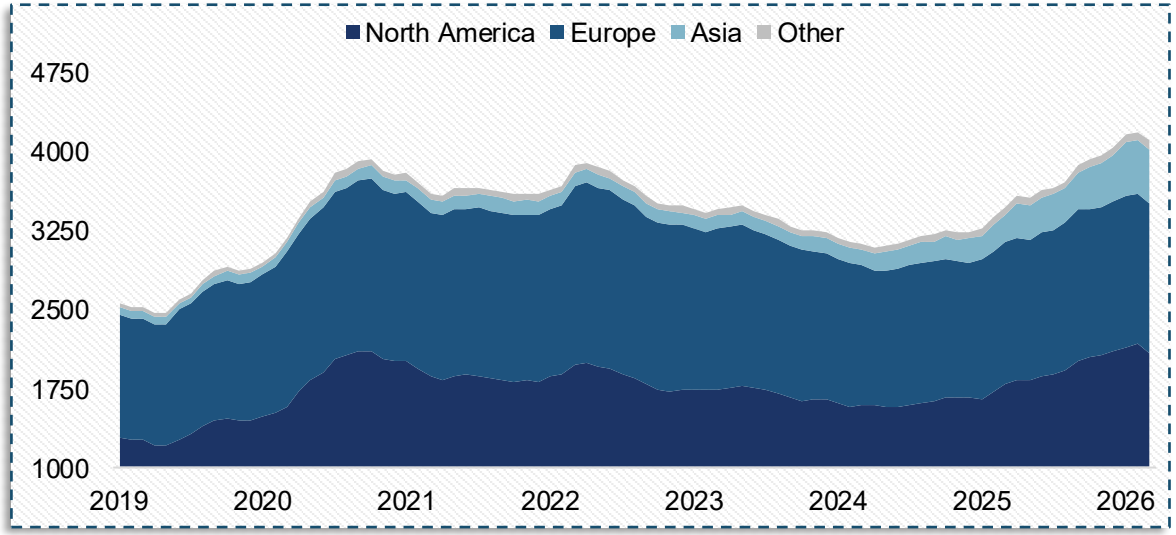
Gold ETF holdings reached the highest level since 2020.

Accumulation

North America leads accumulation, with Europe and Asia adding exposure.

Policy Momentum

Gold ETF inflows accelerate as monetary policy weakens.

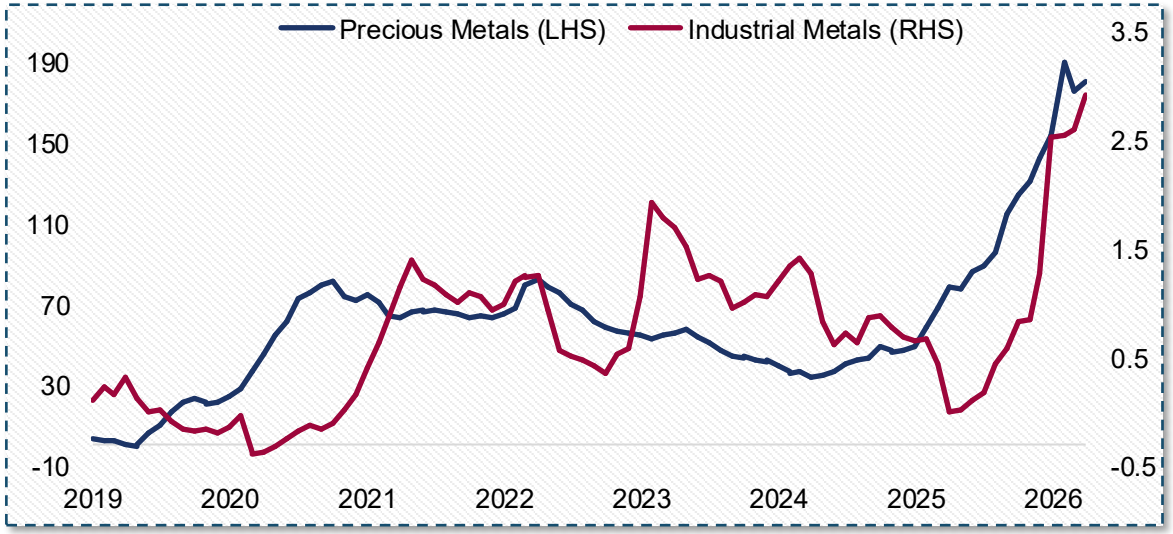


Cumulative ETF Flows: Precious vs. Industrial Metals (Billion USD)

Catalyst
As 60/40 broke down, metals emerged as the go-to alternative.

Beyond Precious
Capital is flowing not only into precious but also into industrial metals.

New Allocation
Investors are increasingly treating metals as a standalone allocation.



Sources: Bloomberg, World Gold Council

© WUTIS – Global Markets

Metals: The Identity Problem

Industrial Commodities with Structural Tailwinds and Cyclical Risk

Productive End-Uses Shape the Economic Sensitivity of Silver and Copper

Silver: Structurally Industrial

~60%

Industry accounts for **>50% of demand**.

~70%

Silver is largely a **by-product**.

5yr

Demand has **out-paced** supply for successive years.

Copper Demand Reality

Construction & Electrical
Over half of global copper demand stems from construction and electrical equipment.

Transport & Machinery
Remaining demand comes from Ev's, industrial machinery and grid technology.

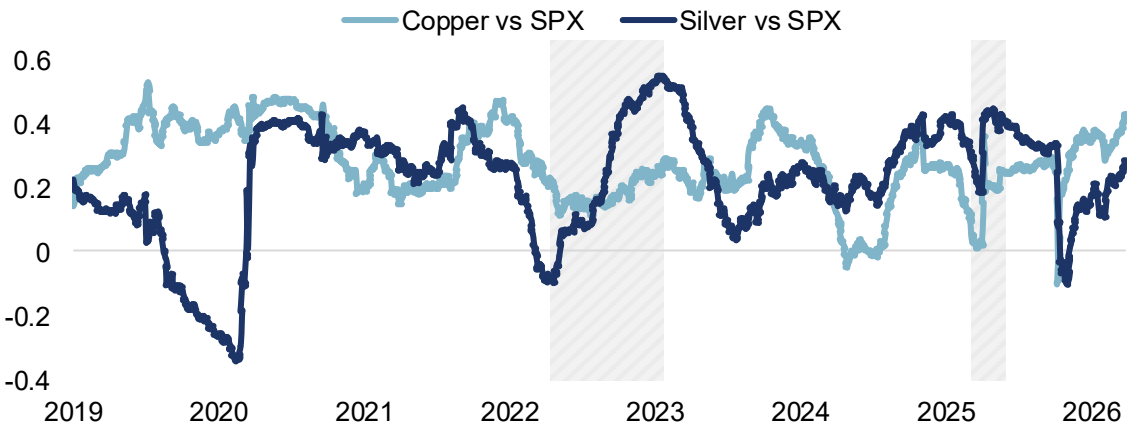
Cyclical Exposure

Industrial demand ties both metals to **manufacturing cycles**.

AI, EV's and grid buildout **deepen** this **dependency** further.

Both commodities therefore **correlate with economic growth**.

6M Rolling Correlation of Daily Returns vs S&P 500



Dependency, Correlation and Contagion

Each investment wave **tightens dependency** on a narrowing set of end-uses.

Neither metal shows a **sustained decorrelation** from equities across the sample.

Deeper dependency shortens the gap between an **idiosyncratic** and a **systematic event**.

Neither metal offers a **structural escape** from equity market dynamics: the same end-uses that drive demand also tie returns to **economic momentum**.

Sources: Bloomberg, Word Silver Survey, IEA

© WUTIS – Global Markets

Fool's Gold

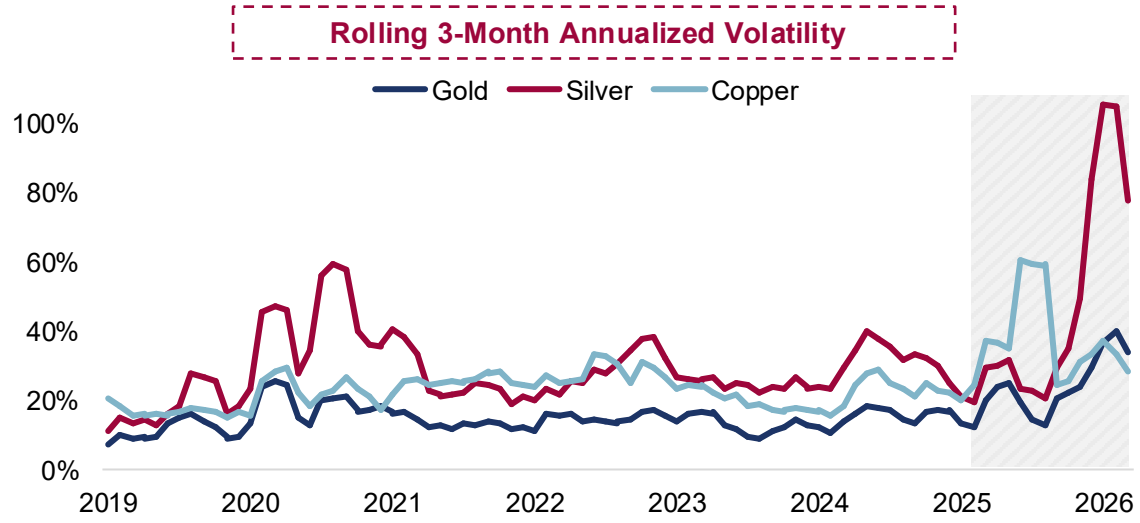
Metals Amplify Risk When Protection is Needed Most

Stress Periods Reveal What Normal Periods Obscure

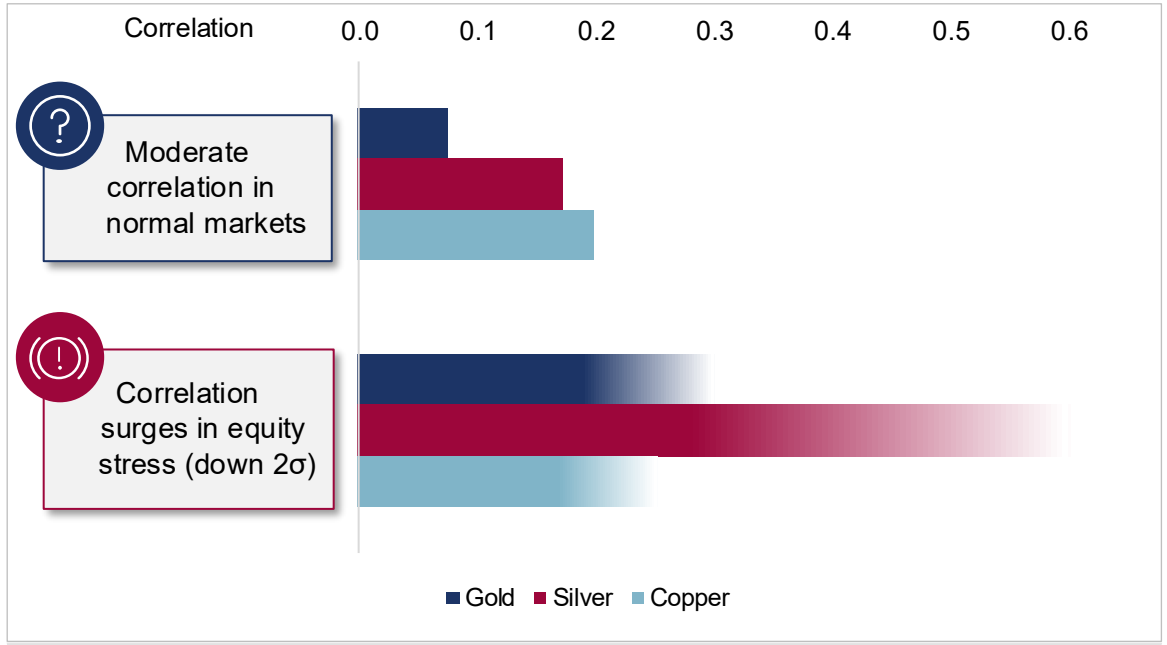
What looks like diversification is a **commodity trade:** the underlying risk profile does not change.

When the S&P draws down by more than 2σ, that **correlation increases across metals**.

The volatility data reinforces this: copper and silver introduce **separate sources of volatility** that compound portfolio risk.



Commodity Correlation With SPX: Normal vs Stress Periods

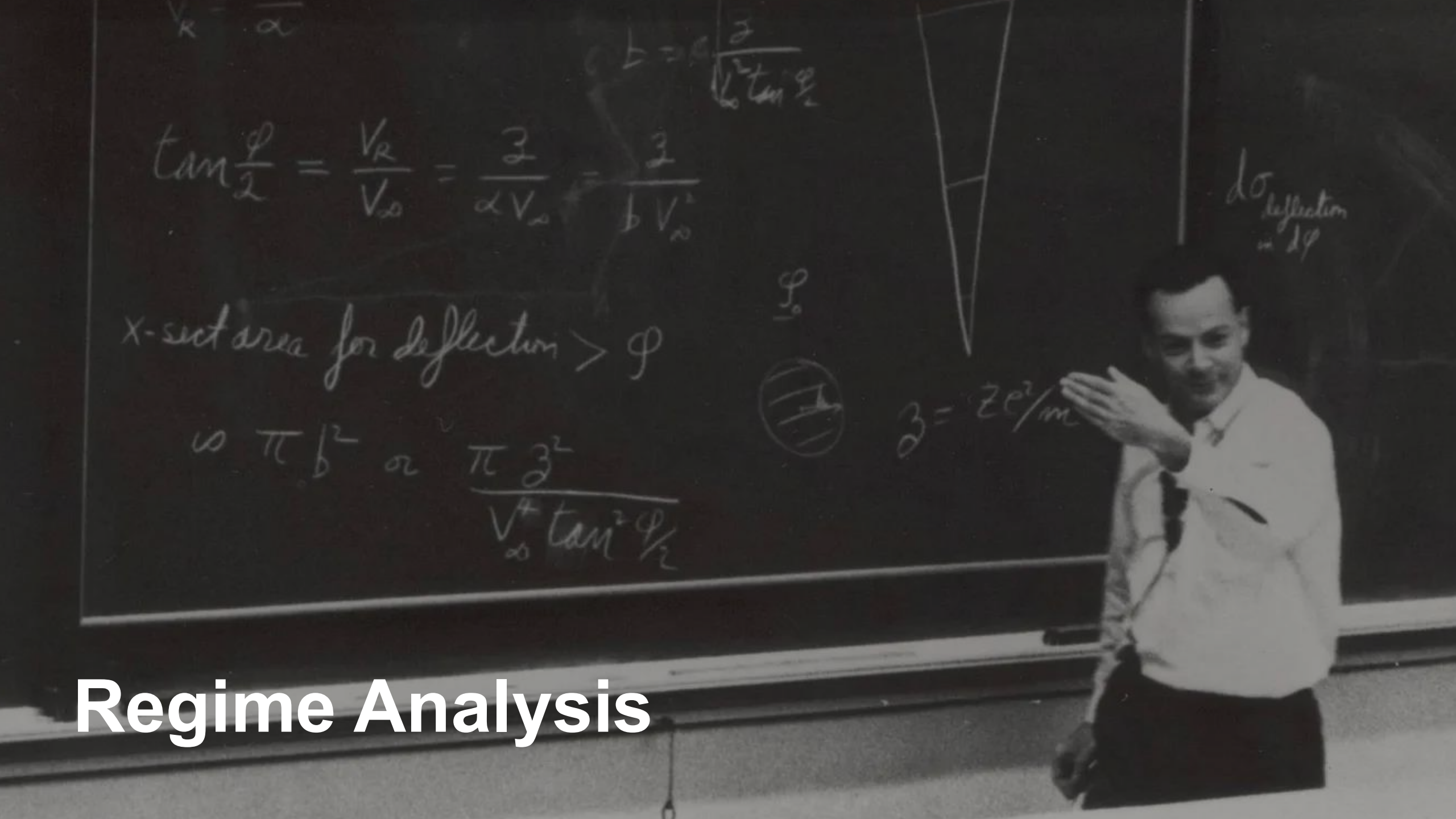


*What if one political decision severs a **critical supply node**, and **turns an isolated shock systematic?***

Commodity dependency ensures that what begins as an idiosyncratic risk **propagates systematically.**

Source: Bloomberg

© WUTIS – Global Markets



Regime Analysis

Regime Analysis I: Hidden Markov Model

Identifying Hidden Regime States Using Quantitative Methods

Model Specification & Empirical Results: 2006-2026

Inputs

1	2	3	4	5
High Yield Spread	**VIX**	**SKEW**	**Industrial Metals**	**Breakeven Inflation**
HY bond risk premium	30-day SPX implied volatility	Implied tail-risk demand	63d LMEX momentum	Implied 10Y inflation

$$\gamma_t(j) = P(q_t = j|O,\lambda) = \frac{a_t(j)\beta_t(j)}{P(O|\lambda)}$$

The model produces a probabilistic regime classification for each day, using forward-backward algorithm.

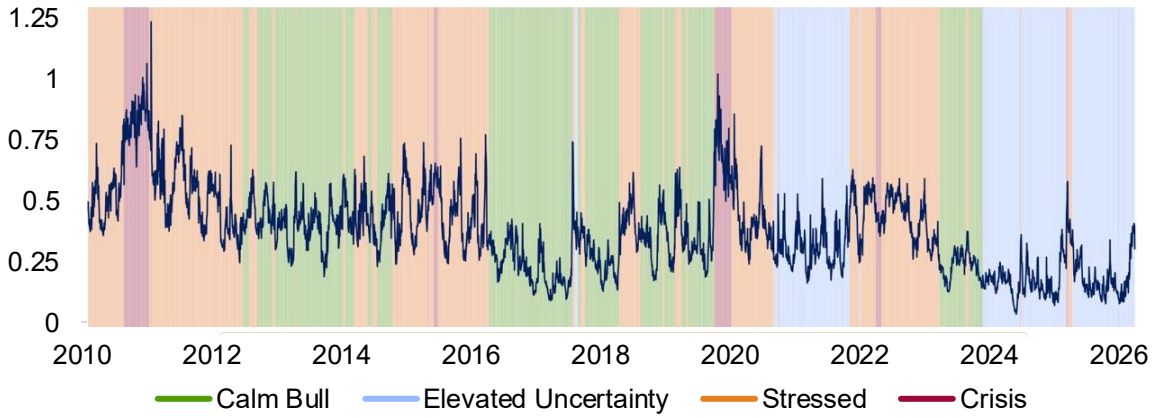
Number of Regimes

4 states were chosen as the fourth state revealed a structurally distinct regime → Elevated Uncertainty.

Model Summary

$\mathcal{N}$	VIX	HY	LMEX	BE10	SKEW	SPX	VVIX	IC
1	13.3	3.79%	0.8%	2.02%	131	22.2%	87.3%	0.3
2	16.8	2.95%	9.6%	2.36%	140	14.1%	98.2%	0.22
3	20.5	5.32%	0.4%	2.09%	124	4.4%	94.1%	0.45
4	37.4	10.1%	-13.9%	1.52%	119	-20.1%	96.3%	0.67

S&P 1M Implied Correlation Index by Regime



Regime Classification

Calm Bull

Highest SPX returns with **lowest volatility**. All five inputs aligned at low risk.

Elevated Uncertainty

Surface looks calm but **credit and options market diverge.** Metals perform the strongest of any regime.

Stressed

The dominant regime. Unified by **elevated stress** without systemic breakdown.

Crisis

All five inputs at extremes simultaneously. The regime captures **economic crashes**.

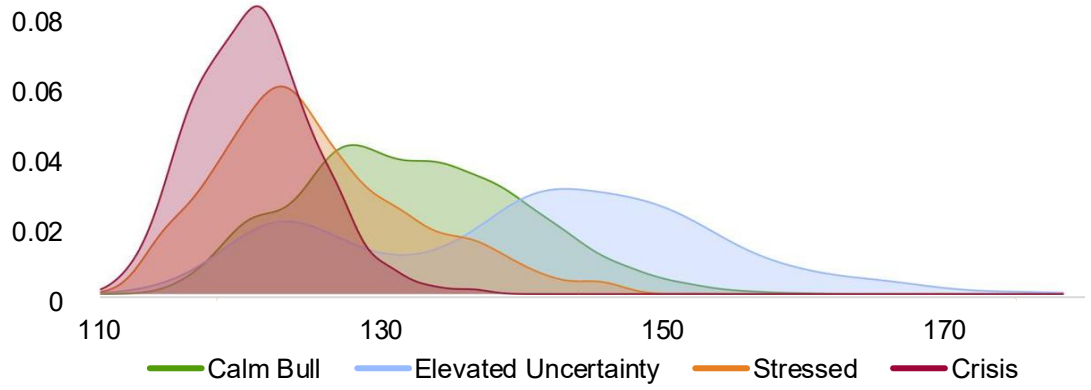
Sources: Bloomberg, Yahoo Finance, Standford University: Hidden Markov Models

© WUTIS – Global Markets

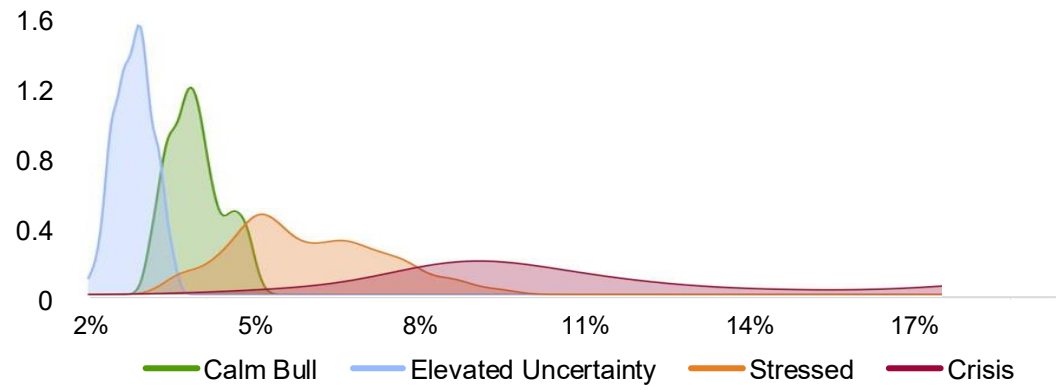
Distribution Analysis I

Compressed Spreads & Elevated Tail-hedging

Distribution of SKEW Index



Distribution of High Yield Spread



Markets hedge with the puts…

Inside Crises **SKEW** collapses to **119**. Once the crash is happening, the implied volatility lifts and **OTM options** become **unaffordable**. New protection-seekers cannot enter.

Elevated Uncertainty has **highest SKEW mean** of all regimes. Bimodal, with half of observations above **140**. The option market is pricing **maximum tail risk**.

The resulting divergence signifies the hedges realization, while **Elevated Uncertainty** clearly shows the **seek for structured product diversification**.

…and less with T-bonds.

Crises share spread-out distribution with **mass concentrated above 7%**, reaching extreme values during GFC. Credit markets price **default risk** and **liquidity freeze** simultaneously.

The **Elevated Uncertainty** does **not share the volatility spill-over** effect from increasing uncertainty, due to the rising government security yields, mitigating the spread.

Markets are not reacting to the risk factors but rather disbelieve the monetary & fiscal defense system. Although **the spreads are tight**, this is **not a sign of healthy environment**.

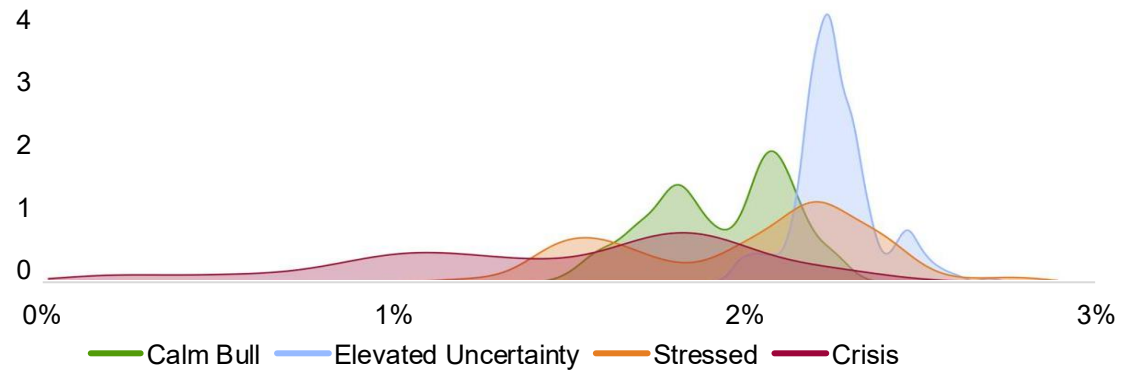
Sources: Bloomberg, Yahoo Finance

© WUTIS – Global Markets

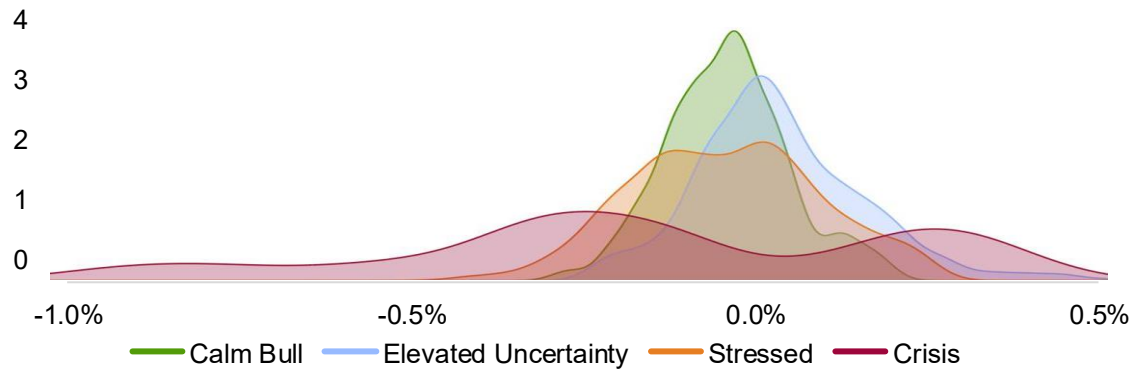
Distribution Analysis II

Pricing Inflation & Driving Metals Momentum

Distribution of 10Y Breakeven Inflation



Distribution of 63d Metals Momentum



Crisis years and unresponsible monetary policy…

Elevated Uncertainty shares tight clustering with **the highest mean** of any regime. This reflects market vulnerability stemming from the **unbalanced monetary policy** and unanchored inflation expectations.

Other market regimes do not form a peak uniformly, although the volatility increases significantly. This outlines the explanatory role of second regime.

…prolonge the alternative bets.

Metals momentum reflects **risky-asset nature** inside the crisis regimes. Metals crash sharply at crisis onset, then recover faster than equities on **fiscal stimulus** and **inventory restocking**.

Elevated Uncertainty mean is the most bullish across other market regimes, with the clearly **defined right-tail distribution**.

Elevated Uncertainty quantitatively reflects the market anticipation of macro-defined crisis and urge to acquire structural diversification.

Sources: Bloomberg, Yahoo Finance

© WUTIS – Global Markets



Trading Strategy

Fundamentals of Dispersion Trading

How Increased Dependency in the Markets Translates into the Volatility Trade

Volatility of Volatility Changed its Shape, Bringing Dispersion Back on the Table

Structured Products changed the mechanics of volatility and hedging.

Macro regime has outlined makret instability and **reshaped anticipation behaviour**.

Rising cross-asset dependencies **merely diversify but heavily elevate** the systematic events.

Lesson from 2026 US-Iran Volatility

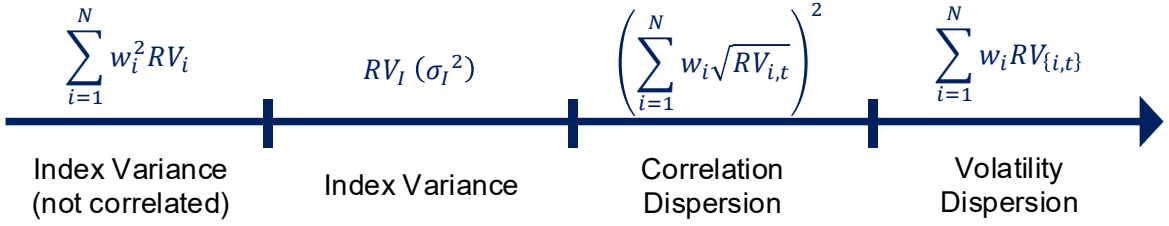
Uncertainty	Crisis	Recovery
Anticipation behaviour led to increased hedging.	Risky assets were left for best, downside volatility compressed.	Compression realized into positive spot-vol beta.

***Takeaway**: There might exist a **room for systematic capturing of convexity mispricing** due to frequent change in sensitivities and fear-adjusting beahviour.*

Mathematical Definition of Dispersion

From Markowitz "Portfolio Selection":

$$\sigma_I^2 = \sum_{i=1}^{N} w_i^2\sigma_i^2 + \sum_{i=1}^{N}\sum_{j=1}^{N} w_i w_j \sigma_i \sigma_j \rho_{\{i,j\}} \quad \ll=\gg \quad \bar{\rho} = \frac{\sigma_I^2 - \sum_{i=1}^{N} w_i^2\sigma_i^2}{\sum_{i=1}^{N}\sum_{j\neq i}^{N} w_i w_j \sigma_i \sigma_j}$$



Approaches to Capture Dispersion

Buying Dispersion: Short Correlation

Trade profits from missing diversification benefit, and stocks exhibiting idiosyncratic uncorrelated momentum.

Selling Dispersion: Long Correlation

Trade profits from systematic correlation of stocks and increase in the composite index variance.

Traditional

ATM Straddle

- ✚ Very liquid and transparent.
- ➖ Dilutes PnL with delta hedging.

- ➖ Illiquid OTC instrument.
- ✚ Pure exposure to the volatility.

Variance Swap

Our Choice

Sources: University of Erlangen-Nürnberg, Societe Generale

© WUTIS – Global Markets

Trading Strategy I

W U T I S

Backtesting Methodology of the Dispersion Trade

Methodology of the Backtest

Month-close volatility and correlation regimes.

Dispersion P&L

$$\sum_{i=1}^{20} N_{var,i}\left(\sigma_{i,real}^2 - K_{i,vol}^2\right) - N_{var,I}\left(\sigma_{I,real}^2 - K_{I,vol}^2\right)$$

Single Stocks | Index

Strategy selection based on the performance in regime t-1.

EOM → BOM → EOM

Modelling Volatility and Correlation Regimes

Volatility Regime

Gaussian Hidden Markov Model (H|M|L) based on following inputs:

Implied Correlation

- Implied 3M50D (Bloomberg Index)
- Terciles (H|M|L)

Realized Correlation

- Realized 1M (Bloomberg Index)
- Terciles (H|M|L)

Daily Log-returns: $ln\left(\frac{P_t}{P_{t-1}}\right)$

SPX Options IV: 3M 50D, 10DP and 10DC

EWMA 5D Half-life:

$$\sqrt{\sum_{i=1}^{N} 0.8 * \left(Ln\left(\frac{h_i}{l_i}\right)^2 - Ln\left(\frac{\max(o_i,c_i)}{\min(o_i,c_i)}\right)^2\right) + 0.2 * Ln\left(\frac{c_i}{c_{i-1}}\right)^2},$$

$$\sigma_i^2 = \lambda * \sigma_{i-1}^2 + (1-\lambda) * x_i^2$$

Pricing Variance Swap Position

$N_{var,i}$: $\frac{\text{Vega Notional}}{2 \text{ x Strike}} = \frac{100\,000}{2 \text{ x } K_{i,var}}$

$K_{var,i}$: $\text{Strike}^2 \approx \sigma_{ATMF}^2 + \beta\sigma_{ATMF}^3 T + \frac{\beta^2}{4}(12\sigma_{ATMF}^2 T + 5\sigma_{ATMF}^4 T^2)$

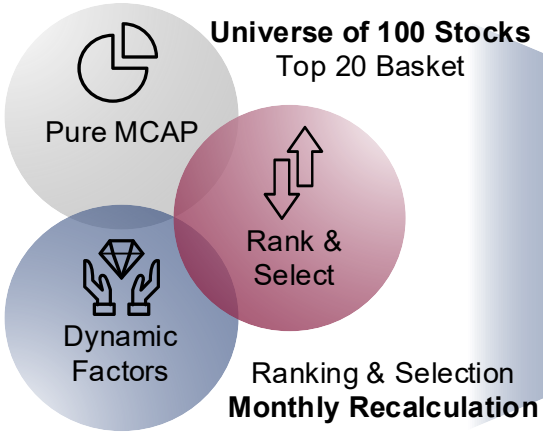
β: $\text{Skew} \approx \frac{\sigma_{90\%} - \sigma_{100\%}}{\ln(0.9)}$

Position is kept **vega-neutral** at inception to avoid parallel move sensitivity.

Position is open-closed on monthly basis, based on **fair strike and log-variance**.

No interim performance calculation, due to the **interpolation complexity**.

Smart-Dirty Dispersion



Universe of 100 Stocks
Top 20 Basket

Ranking & Selection
Monthly Recalculation

Implementation

Jan. 2010 - Jan. 2022:
- Monthly P&L for each approach.
- Group regime by mean P&L across strategies.

After Jan. 2022:
- Calculate last month regime.
- Rank mean regimes by mean P&L.
- Select best performer based on last regime.

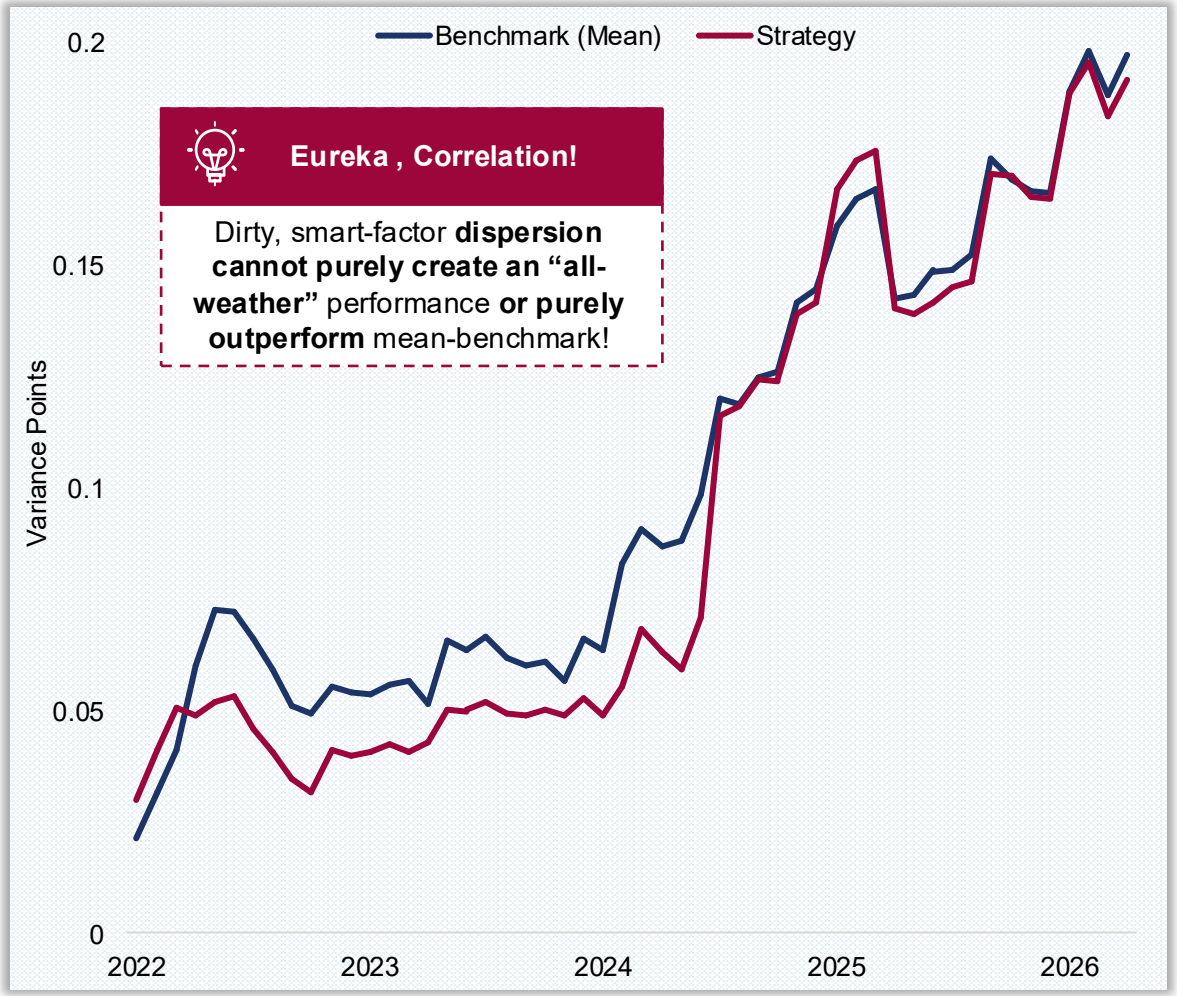
Sources: JPMorgan, Societe Generale, Bloomberg, Goldman Sachs

© WUTIS – Global Markets

Trading Strategy II

Out-of-sample Testing and Trade Structure Implementation

Cumulative Performance of the Best-Mean-Dispersion (2022 – 2026)



Research Conclusion: Empirical Role of Volatility & Correlation

Dispersion is a volatility selling approach, grinding higher, while being exposed to **the convex events**.

Strategies aiming to capture the "correlation"-trade from historic dispersion performance are struggling to survive due to convexity **bleed** or are sample-biased.

Discretionary expertise has a very **strong potential** inside the dispersion trading, offering convex convexity, although costs should be considered.

Trade Structure: May 2026

Model Selection

Long the highest **EPS 12M Growth Dispersion** form the universe.

Discretionary Consideration

Earnings season after the geopolitical cease open the **room for single name dispersion**.

Short Leg

Palantir, Disney+, MicroStrategy, Lilly, Johnson & Johnson

Total Variance Notional: 29,179.79

Long Leg

S&P 500®

Variance Notional: 65,582.30

© WUTIS – Global Markets



Appendix

Appendix

Sources

Source Name	Link	Date of Retrieval	Used for
Bloomberg	-	02.05.2026	Continuous Work
Yahoo Finance API	-	02.05.2026	Continuous Work
Standard & Poor: Capital IQ	Link	01.05.2026	Stock Filtering, Factor Selection
University of Erlangen-Nürnberg (Statistics and Econometrics Dept.)	Link	24.04.2026	Dispersion Mathematics
Societe Generale	“A regime-aware factor framework for smarter dispersion trades”	24.04.2026	Volatility and Regimes
Societe Generale	“How to hedge a slow equity selloff”	24.04.2026	Hidden Markov Model (Dispersion)
Societe Generale	“Vol targeting in practice: What works and what doesn’t”	24.04.2026	EWMA Formula
JPMogan Chase & Co.	Link	20.04.2026	Pricing of Variance Swaps
Goldman Sachs	Link	20.04.2026	Pricing of Variance Swaps
Structured Retail Products	Link	18.04.2026	Structured Products Dynamics
Barclays Derivatives Research	“Volatility Outlook 2026: Tech Support for Tails”	18.04.2026	Structured Products Dynamics

© WUTIS – Global Markets

Appendix

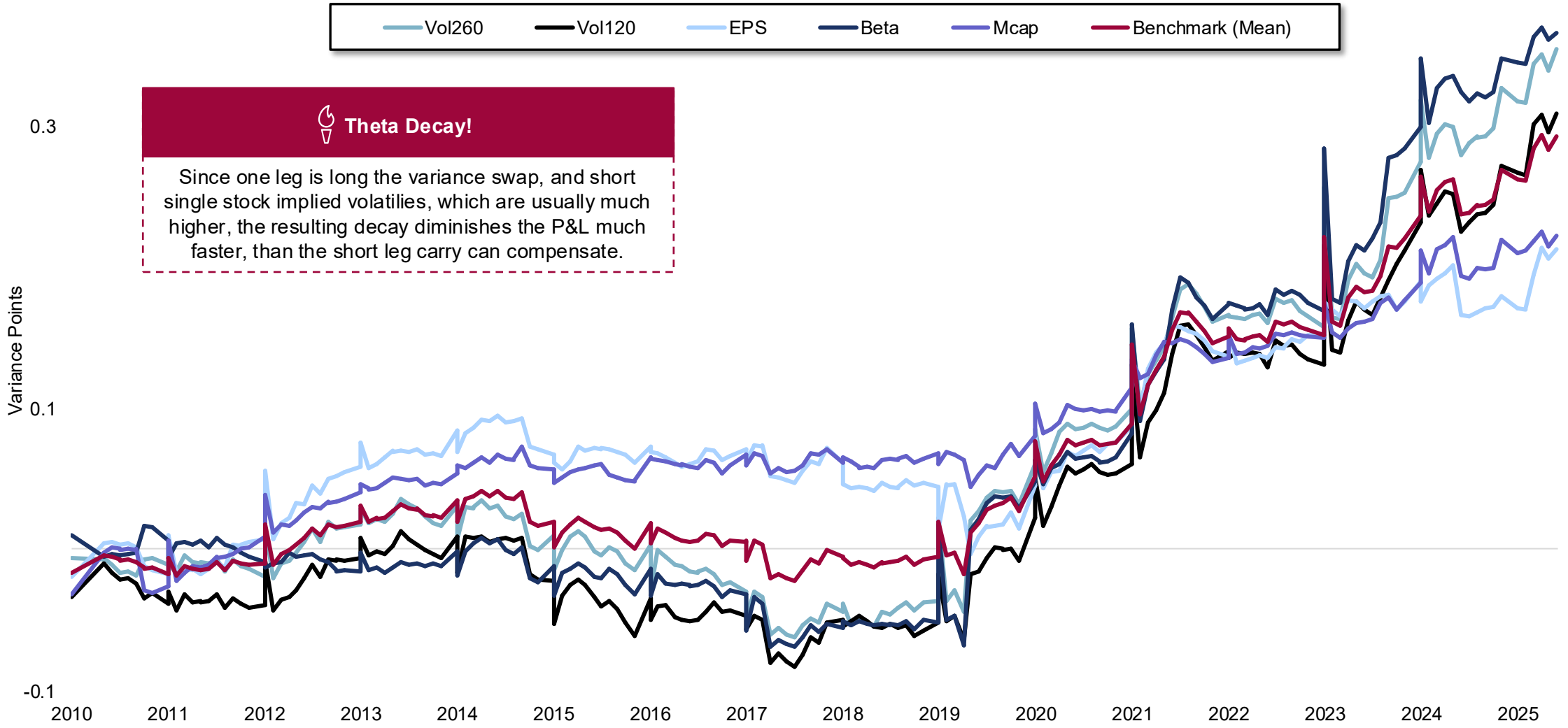
Sources

Source Name	Link	Date of Retrieval	Used for
World Gold Council	Link	19.04.2026	Gold ETF Holdings
The Silver Institute	Link	22.04.2026	Silver Demand Dynamics
International Energy Agency	Link	22.04.2026	Copper and Industrial Metals Demand
Federal Reserve Bank of St. Louis	Link	20.04.2026	Market and Correlation Analysis
Standford University: Hidden Markov Models	Link	15.04.2026	Hidden Markov Model

© WUTIS – Global Markets

Appendix

Performance of Dirty Dispersion Strategies Over the Years



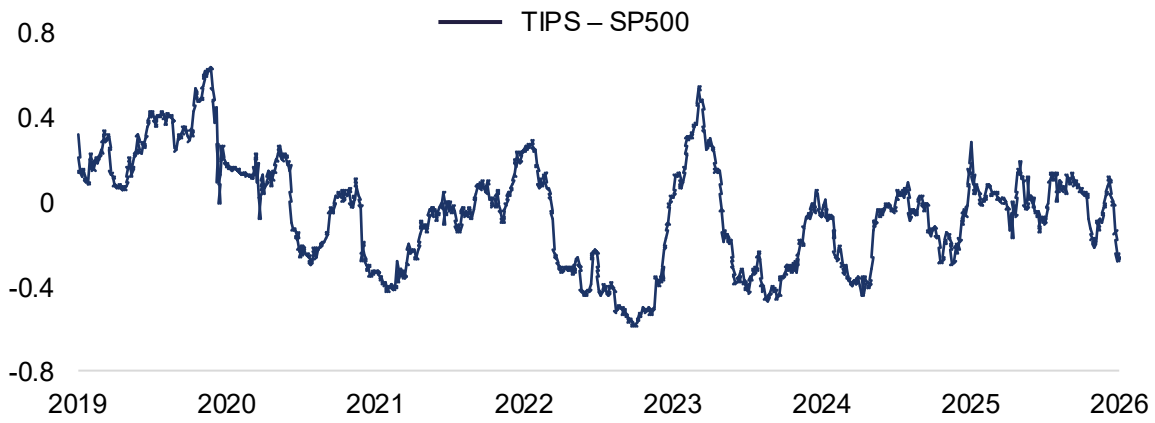
Sources: Bloomberg, Inhouse Calculation

© WUTIS – Global Markets

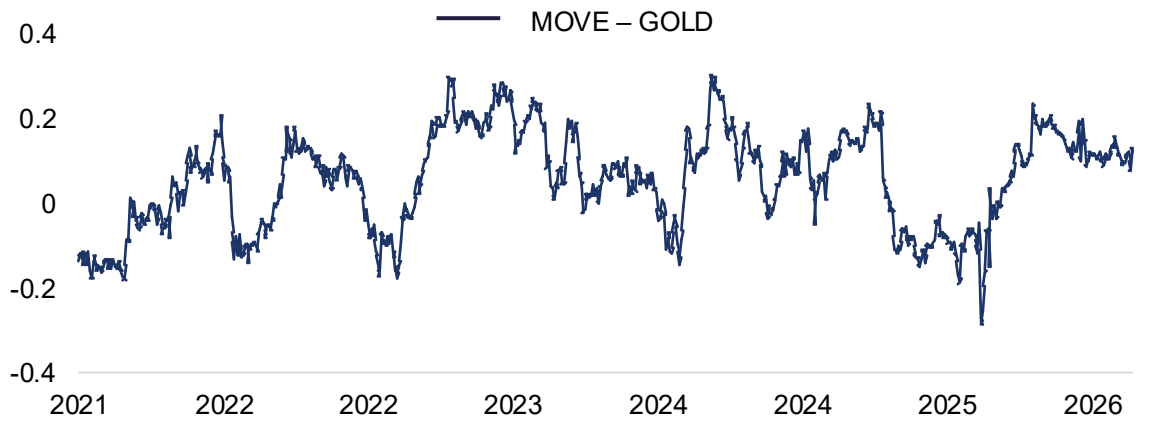
Correlation Analysis

Macro Regimes Driving Cross-Asset Correlations: 3M-Rolling

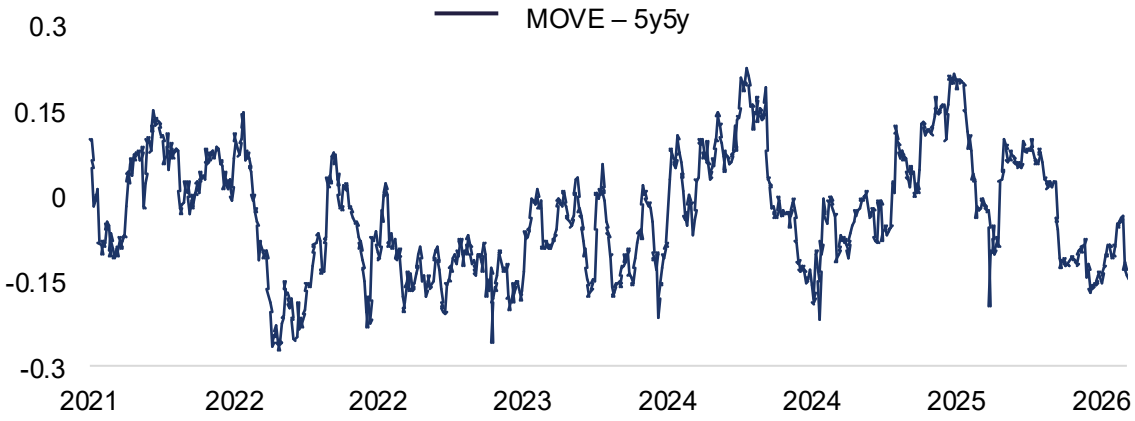
10-year Real Yields vs S&P 500 Returns



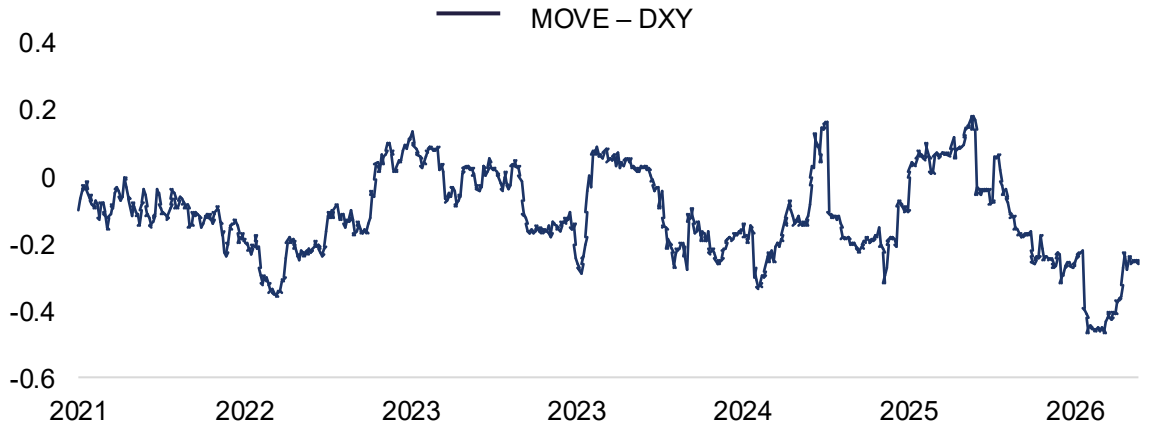
MOVE Index and Gold Returns



MOVE Index and 5Y5Y Inflation Correlation



MOVE Index and US Dollar



Sources: FRED, Yahoo Finance

© WUTIS – Global Markets

Disclaimer

This document is being made available by WUTIS Trading and Investment Society, acting through its society ("WUTIS"), for information purposes only and for the sole and exclusive use of the addressee in connection with the matter or possible transaction to which this document relates. This document is incomplete without reference to, and should be assessed solely in conjunction with, the oral briefing provided by WUTIS.

This document is strictly confidential and remains the property of WUTIS. You agree that you shall only use this document for the purpose of evaluating and considering the matter or possible transaction to which this document relates. This document may not be distributed, published, reproduced, or disclosed, in whole or in part, to any other person, nor relied upon by you or any other person, nor used for any other purpose at any time, in each case without the prior written consent of WUTIS.

This document does not constitute nor does it form part of an offer to sell or purchase, or the solicitation of an offer to sell or purchase, any securities or any of the businesses or assets described herein or an offer or recommendation to enter into any transaction described herein nor does this document constitute an offer or commitment to provide, arrange or underwrite any financing. You are responsible for making your own independent investigation and appraisal of the risks, benefits, appropriateness and suitability of any transaction or matter contemplated by this document and WUTIS is not making any recommendation (personal or otherwise) or giving any investment advice and will have no liability with respect thereto. The decision to proceed with any transaction or action contemplated by this document must be made by you in the light of your own commercial assessments and WUTIS will not be responsible for such assessments.

Neither WUTIS nor any of its subsidiaries or affiliates, nor any of their respective officers, directors, employees or agents (WUTIS together with such persons being the “WUTIS Group”) accepts any liability whatsoever for any direct, indirect or consequential losses (in contract, tort or otherwise) arising from the use of this document or its contents or any reliance on the information contained herein. WUTIS Group is not responsible for any specialist advice, including legal, regulatory, accounting, model accounting, tax, actuarial or other advice.

This document was prepared on the basis of information and data, obtained from publicly available sources and, where applicable, from you and/or any other entity that may be involved in any transaction or matter contemplated by this document (and/or any of your or the aforementioned entities’ affiliates), in each case prior to or on the date hereof. The information in this document has not been independently verified by WUTIS. WUTIS has relied on any information provided by you or from third party or public sources as complete, true, fair, accurate and not misleading. With respect to any financial or operating forecasts and analyses provided to it, WUTIS has assumed that they are achievable and have been reasonably and properly prepared on bases reflecting the best currently available information, estimates and judgments as to the future financial performance of the entity(ies) to which they relate and that such forecasts or analyses would be realised in the amounts and time periods contemplated thereby. No member of the WUTIS Group, makes any warranty or representation, express or implied, as to the accuracy, completeness or reasonableness of the information (including projections and assumptions) contained in this document whether obtained from or based upon third party or public sources or otherwise.

This document is given as at the date hereof, may not be final, is based on information available to WUTIS as at the date hereof, is subject to any assumptions set out therein and is subject to change without notice. Accordingly, this document may be based on (a) data and information that may no longer be current and (b) estimates that may involve highly subjective assessments. It should be understood that subsequent developments may affect this document and the WUTIS Group does not undertake any obligation to provide any additional information or to update any of the information or the conclusions contained herein or to correct any inaccuracies which may become apparent. Any data or information regarding, or based on, past performance is no indication of future performance.

Members of the WUTIS Group are involved in a wide range of commercial banking, investment banking and other activities out of which conflicting interests or duties may arise. In the ordinary course of its business, the WUTIS Group may provide services to any other entity or person whether or not a member of the same group as you (a “Third Party”), engage in any transaction (whether on its own account, on behalf of any Third Party or otherwise, and including any transaction or matter contemplated by this document), notwithstanding that such services, transactions or actions may be adverse to you or any member of your group, and the WUTIS Group may retain for its own benefit any related remuneration or profit. The WUTIS Group operates in accordance with a conflicts of interest policy which identifies conflicts of interest it faces in the ordinary course of its business, and establishes organisational and procedural measures to manage those conflicts where it is reasonably able to do so. Neither WUTIS nor any other part of the WUTIS Group shall have any duty to disclose to you or utilise for your benefit any non-public information acquired in the course of providing services to any other person, engaging in any transaction (on its own account or otherwise) or otherwise carrying on its business. WUTIS’ research analysts and research departments are independent and are subject to certain regulations and internal policies. WUTIS research analysts may hold opinions and make statements or investment recommendations and/or publish research reports with respect to any company referred to herein, the transactions contemplated herein or any person or entity involved therein or related thereto that differ from or are inconsistent with the views or advice communicated by WUTIS.

© WUTIS – Global Markets